Venkata Lakshmi Parimala Pasupuleti

Model Validation Specialist · AML · Fraud · Financial Crime Compliance · AI/ML Model Validation · Python · SQL

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Analyst with 7+ years in **financial crime compliance and fraud detection** combined with hands-on **AI/ML model validation** — exactly the combination SMBC seeks. At Entrata, spent 7+ years as the human validation layer against income fraud and household-composition fraud across **1.6M+ HUD/Tax Credit financial records**: reviewing income documentation, applying LIHTC/Section 42 asset limits, flagging AML-adjacent inconsistencies, and verifying household declarations — with **100% PHI compliance, zero violations**. At Intuit, now builds **AI/ML model validation frameworks** (LLM-as-judge + golden datasets) assessing **conceptual soundness, data quality, performance monitoring, and outcome assessment**; compiles validation reports with structured audit trails for stakeholder review. AgentProbe, my open-source eval toolkit, scores model outputs on accuracy, hallucination, retrieval quality, completeness, and **compliance** across 5 LLM providers. LeaseI (8-agent LangGraph) achieved **93.6% precision, 87.9% F1** on financial compliance validation. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Model Validation & Financial Crime: AML/Fraud Model Validation, Financial Crime Compliance, AI/ML Model Validation, Conceptual Soundness, Data Quality Assessment, Benchmarking, Sensitivity Analysis, Back-testing, Outcome Assessment, Model Governance, MRM Standards, Model Risk Documentation & Audit Trails

Python & ML: Python (9+ yrs), LangGraph, LangChain, FastAPI, scikit-learn, XGBoost, Pandas, NumPy, HuggingFace, spaCy, BERT, ChromaDB

Data & Financial Records: SQL, Snowflake, PostgreSQL, SQLite, ETL Pipelines, Data Quality Monitoring, 1.6M+ HUD/Tax Credit Records, PHI/HIPAA Compliance, LIHTC/HCV/HUD Regulatory Workflows

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), GCP/Vertex AI, Docker, GitHub Actions CI/CD, Git, Linux

WORK EXPERIENCE

Intuit

Jun 2026 – Present

Software Engineer — AI/ML Model Validation & Financial Analytics

- **Build and execute AI/ML model validation frameworks** for financial analytics: LLM-as-judge evaluation suites + golden datasets assessing **model conceptual soundness, data quality, performance monitoring, and outcome assessment**; design **benchmarking, sensitivity analysis, and back-testing** approaches to evaluate model effectiveness and fitness for purpose; compile model validation reports with **structured audit trails** identifying risks and escalating findings to stakeholders.
- Maintain **model risk governance documentation** aligned with internal AI governance standards; add guardrails around model outputs (input validation, sensitive-data handling, fallbacks); collaborate with data science teams to strengthen model risk management practices across the enterprise.

Entrata — Multifamily & Affordable Housing SaaS

Jan 2017 – Jul 2024

Analytic Software Engineer — Financial Crime Compliance & Fraud Detection

- **Operated as the validation layer for financial crime and income fraud** across 1.6M+ HUD/Tax Credit records: reviewed income documentation, detected fake paystubs and altered bank statements, applied LIHTC/Section 42 income and asset limits, flagged household-composition discrepancies, and verified asset disclosures — within fair-housing constraints requiring uniform, document-driven verification. **100% PHI compliance, zero violations** across 7+ years.
- Encoded manual fraud detection judgment into automated Snowflake-backed pipelines — eliminating **80% of manual review workload** with zero compliance violations; optimized **200+ SQL queries (30% performance improvement)**; managed data access governance via AWS S3 Libsodium-encrypted pipelines with role-based access controls; delivered across **120+ Agile sprints**.

PROJECTS

AgentProbe — AI/ML Model Validation & Evaluation Toolkit

Model Validation · Compliance

Python · FastAPI · LangGraph · ChromaDB · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Open-source **AI/ML model validation toolkit**: 5-metric evaluation engine (accuracy, hallucination rate, retrieval quality, completeness, compliance) across 5 LLM providers; **benchmarking and outcome assessment** via 3 LangGraph agents and CI golden-dataset regression tests; 15-endpoint FastAPI backend for production-grade validation workflows.

Leasel — Financial Crime & Compliance Validation Pipeline

Fraud · Compliance Audit

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG · scikit-learn · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end **financial compliance validation pipeline** over 1,240 lease PDFs: feature extraction, rule-based classification, deterministic audit engine with full metadata and lineage; **LLM-as-judge evaluation: 93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready. Core design: LLM reasons, deterministic logic makes every regulated call.

CrashClarity — Predictive Risk Modeling on 104K Records

ML · Risk Analytics

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering

- Random Forest + Logistic Regression on 104K+ records; **sensitivity analysis, back-testing, and outcome assessment**; **85%+ classification accuracy**; self-service dashboard for stakeholder reporting.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML pipelines, data governance, anomaly detection, financial data analysis

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017